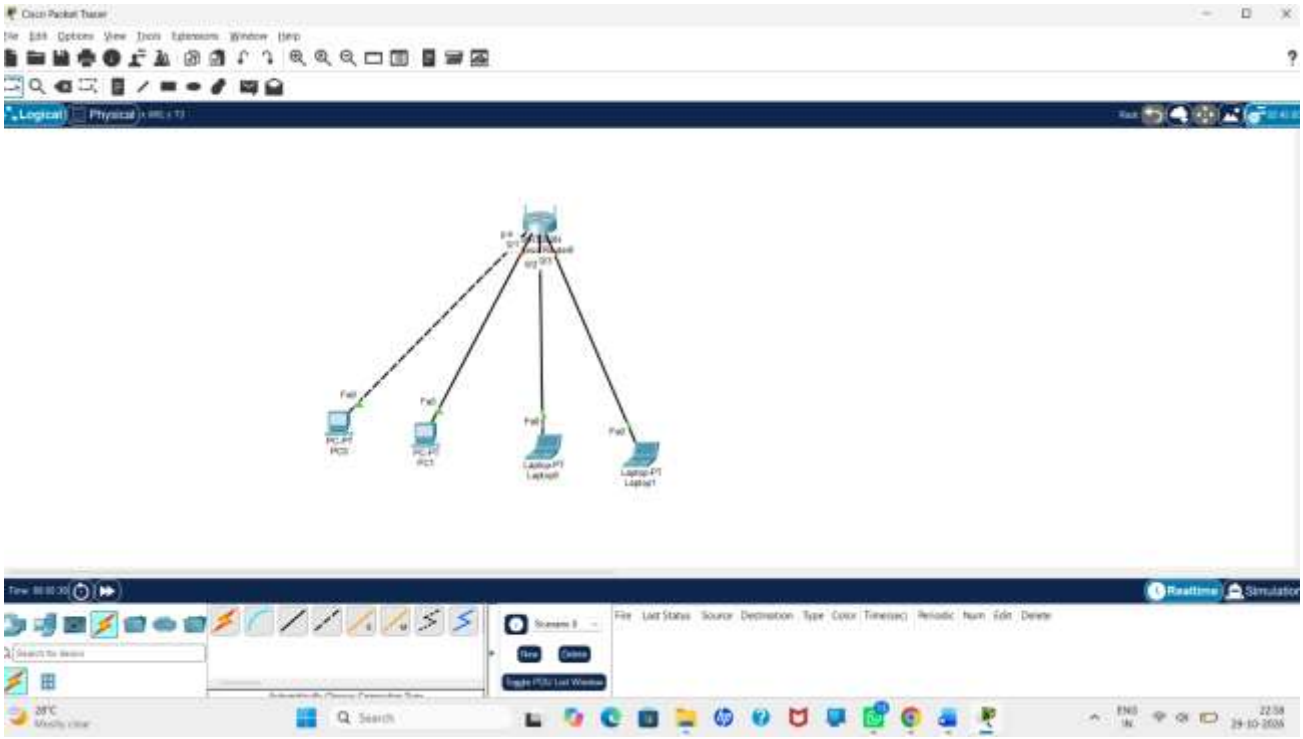
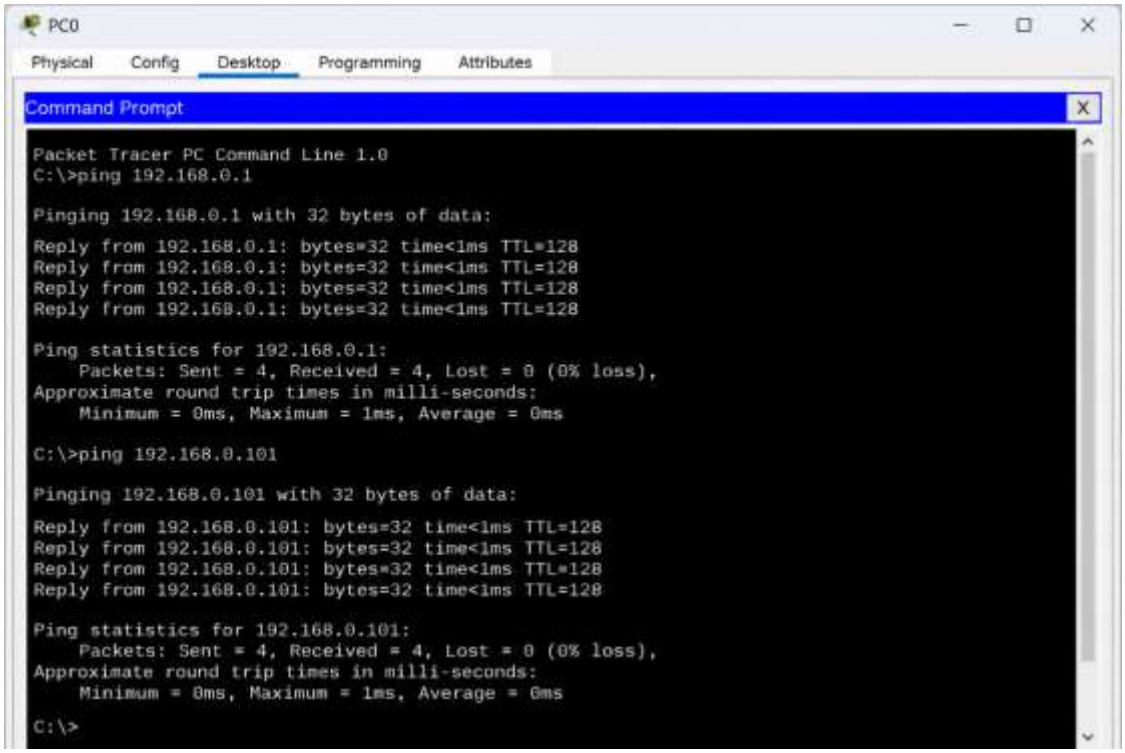
Review and Practice Practical

Q1. Open Cisco Packet Tracer and create a new project file named 'Home WiFi Network.pkt' with three PCs and one wireless router connected appropriately.

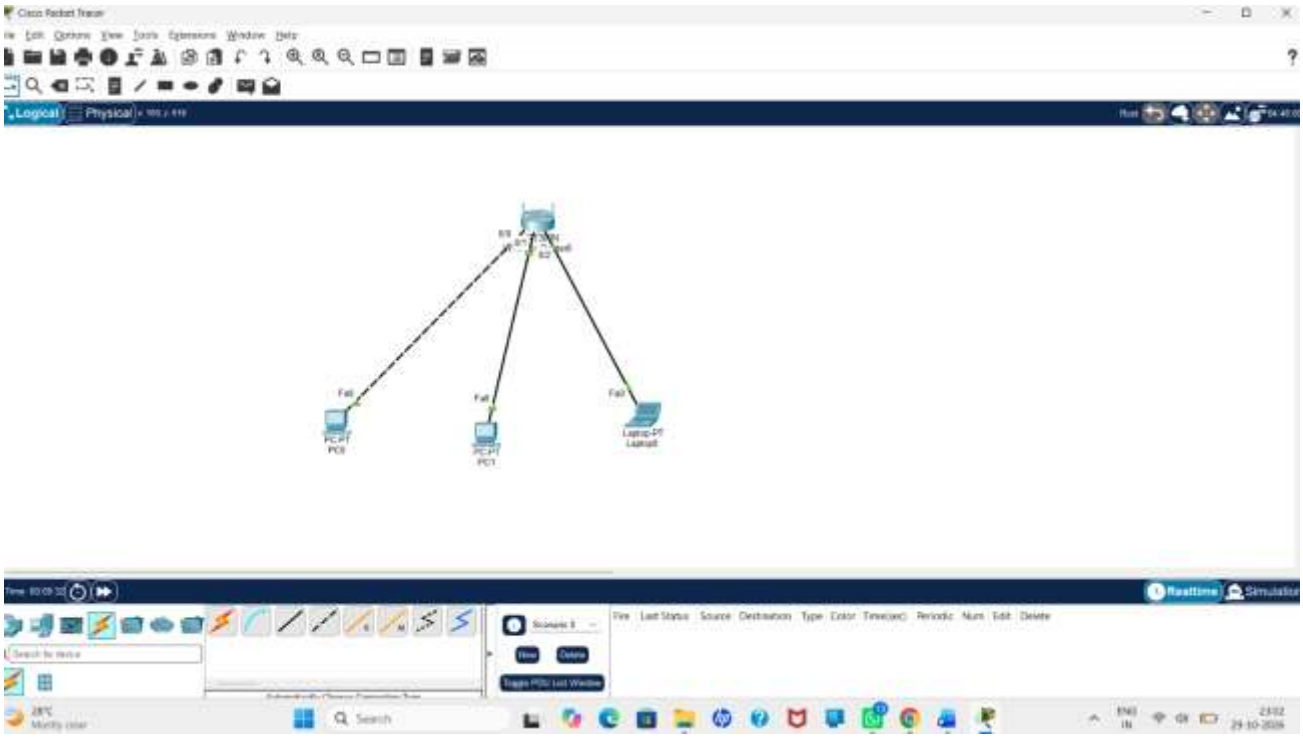
Aim

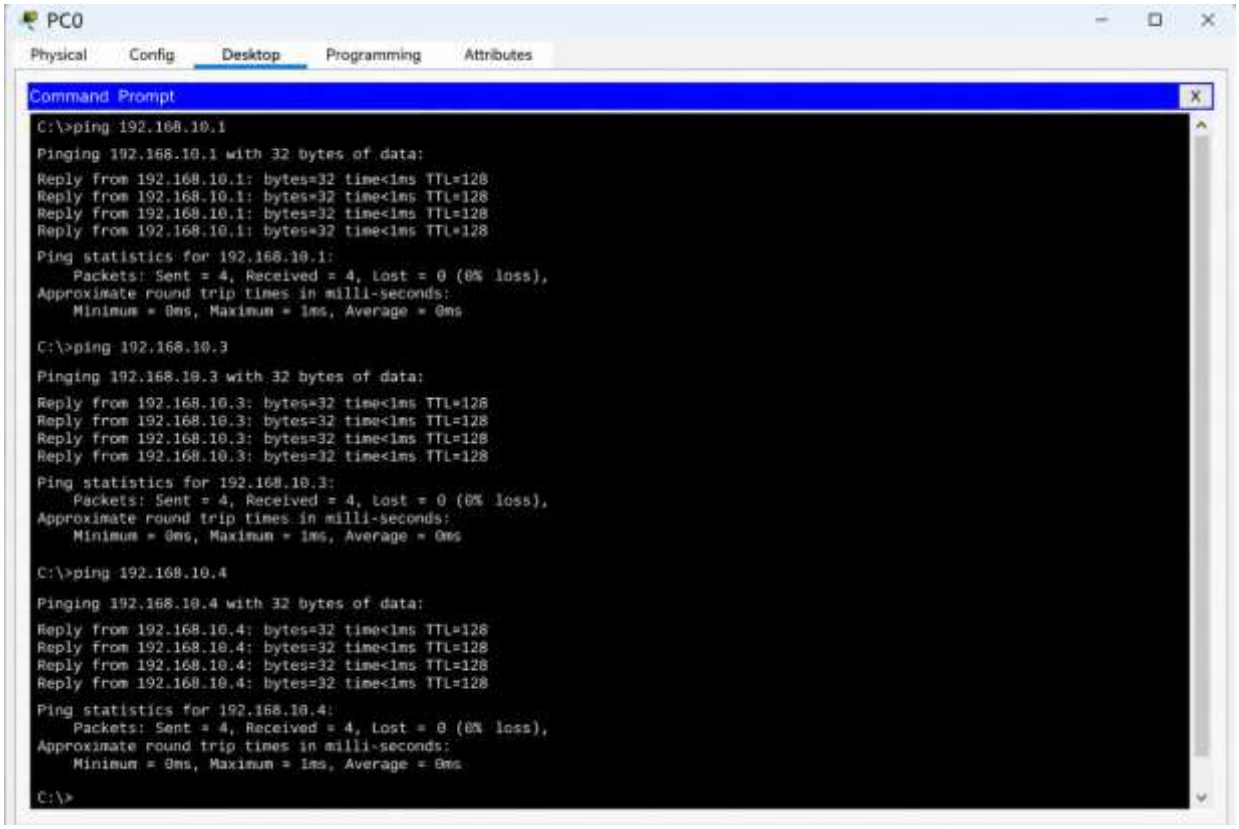
To create a home network using three PCs and one wireless router in Cisco Packet Tracer, configure IP addresses using DHCP, test connectivity, and save the project.





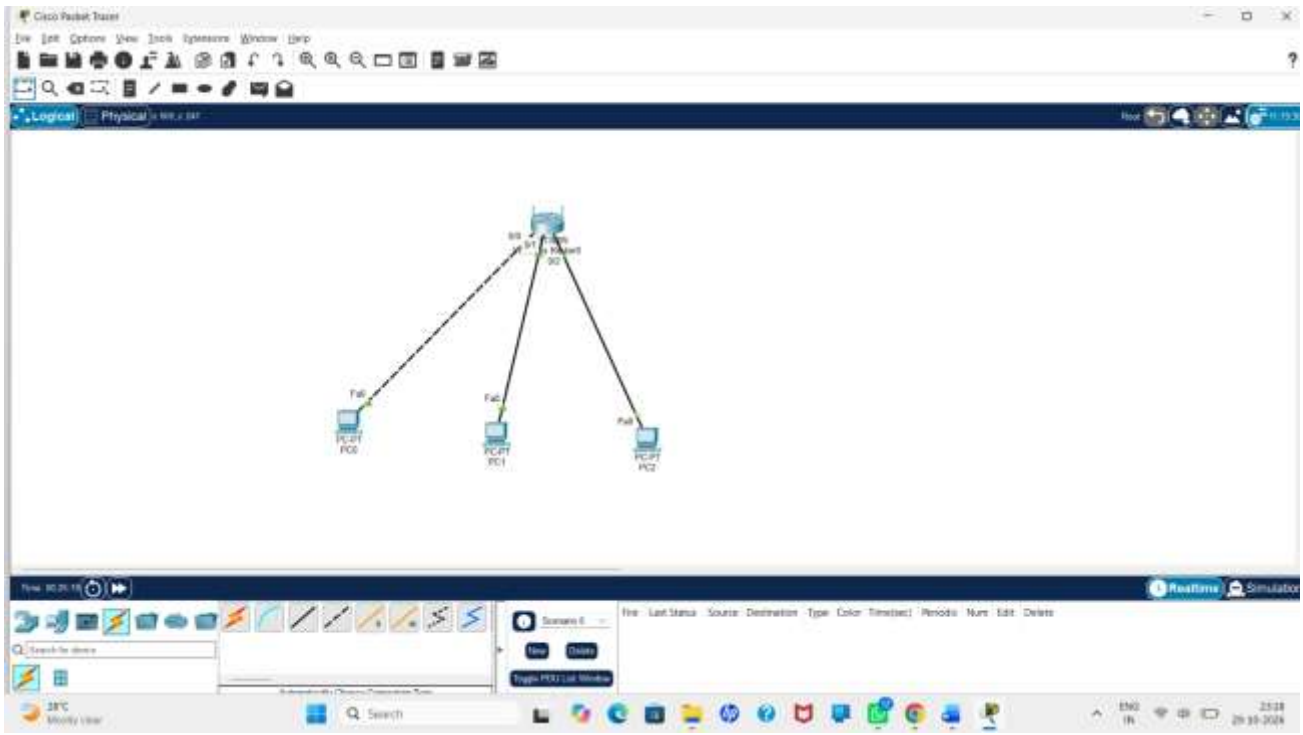
Q2 Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

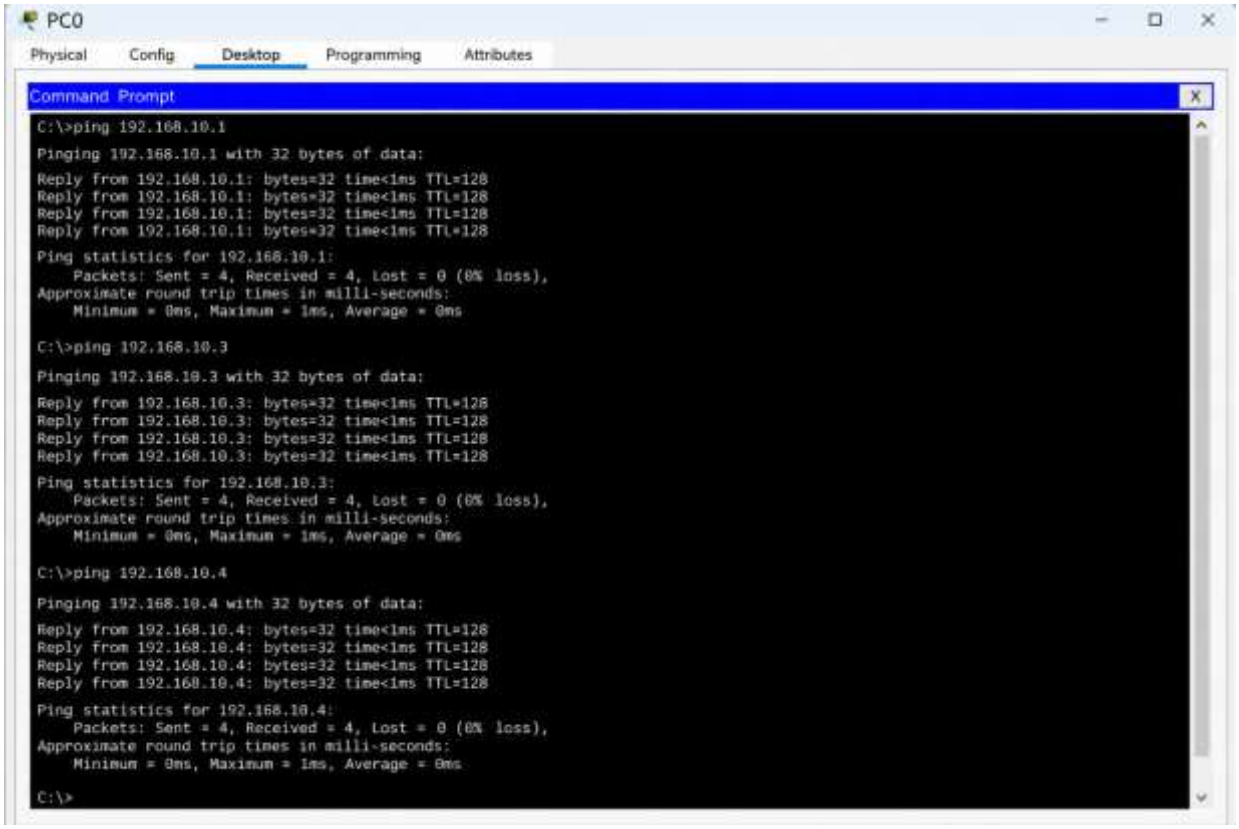




**Q3. Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router.

Hint: Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'.**





Q4 Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

Home Wi-Fi Network Configuration
Cisco Packet Tracer Practical Report | Project: HomeWiFiNetwork.pkt
Aim: Build a network with three PCs and one WRT300N wireless router, assign static IPv4 addresses, and verify connectivity using ping.
1. Required Devices
3 × PC-PT, 1 × WRT300N Wireless Router, 3 × Copper Straight-Through cables.
2. Build the Network Topology
Open Cisco Packet Tracer and choose File > New.
Place three PC-PT devices from End Devices in the workspace.
Place one WRT300N from Network Devices > Wireless Devices.
Connect PC0 FastEthernet0 to Ethernet 1, PC1 FastEthernet0 to Ethernet 2, and PC2 FastEthernet0 to Ethernet 3 using straight-through cables.
Wait for link indicators to turn green.

Illustrative Packet Tracer-style image (not an actual project screenshot).
3. Configure the Router
Click the WRT300N router and select GUI > Setup > Basic Setup.
Set Local IP Address to 192.168.10.1 and Subnet Mask to 255.255.255.0.
Click Save Settings. The DHCP server may remain enabled provided its pool does not overlap the manually assigned PC addresses (for example, start at 192.168.10.100).

Illustrative Packet Tracer-style image (not an actual project screenshot).
4. Configure Static IP Addresses
On each PC open Desktop > IP Configuration, choose Static, and enter the values below.

Device	IPv4 Address	Subnet Mask	Default Gateway

PC0	**192.168.10.2**	**255.255.255.0**	**192.168.10.1**
PC1	**192.168.10.3**	**255.255.255.0**	**192.168.10.1**
PC2	**192.168.10.4**	**255.255.255.0**	**192.168.10.1**

PC0 — IP Configuration

Illustrative Packet Tracer-style image (not an actual project screenshot).

PC1 — IP Configuration

Illustrative Packet Tracer-style image (not an actual project screenshot).

PC2 — IP Configuration

Illustrative Packet Tracer-style image (not an actual project screenshot).

5. Verify Configuration and Test Ping

Open Desktop > Command Prompt on each PC. Run ipconfig to confirm its IP address, then run the following commands:

PC0

ping 192.168.10.1

ping 192.168.10.3

ping 192.168.10.4

PC1

ping 192.168.10.1

ping 192.168.10.2

ping 192.168.10.4

PC2

ping 192.168.10.1

ping 192.168.10.2

ping 192.168.10.3

A successful test shows Reply from ... and typically Sent = 4, Received = 4, Lost = 0 (0% loss). Initial ARP resolution can occasionally cause a first ping to time out; repeat the test if necessary.

The following is an illustrative ping output, not evidence of an executed test:

6. Save the Project

Click File > Save As.

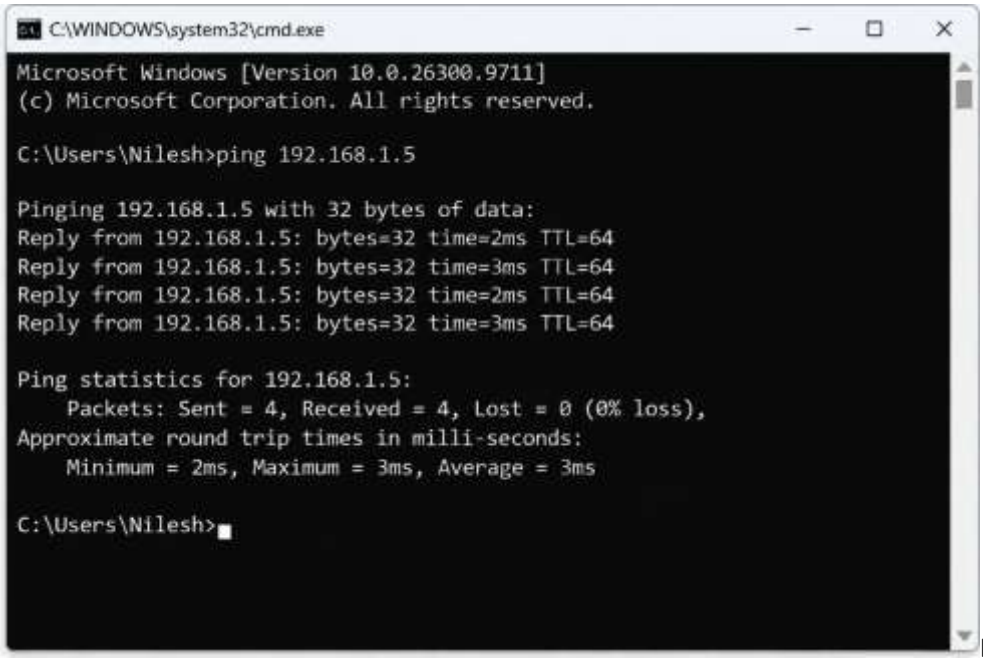
Name the project HomeWiFiNetwork.pkt and save it in your assignment folder.

Capture real screenshots of your topology, PC IP Configuration windows, and ping results for submission.

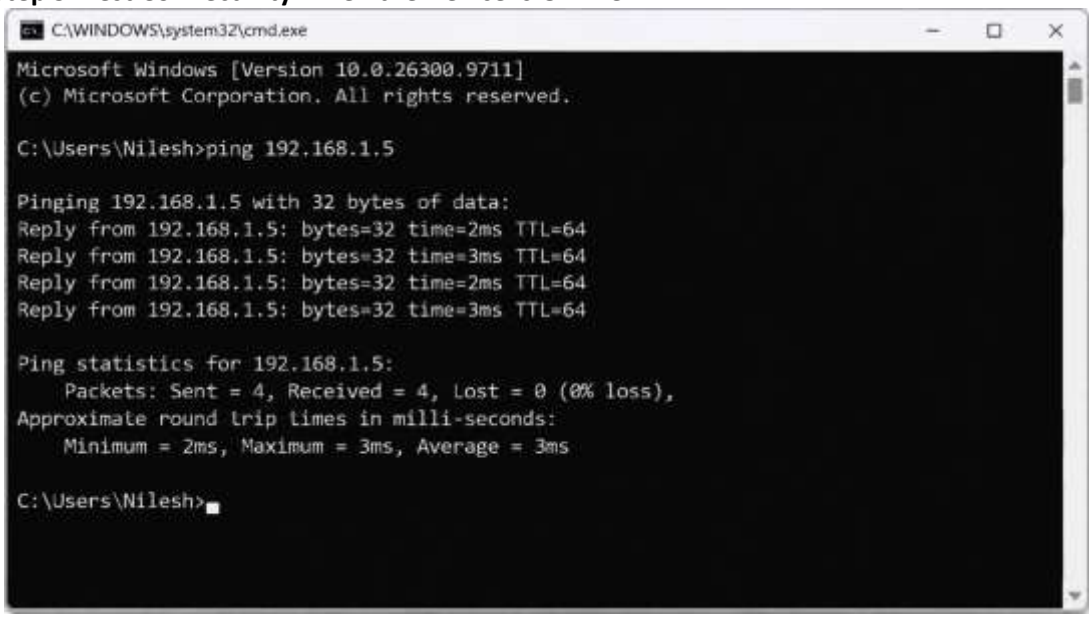
7. Result and Conclusion

The intended configuration places PC0, PC1, PC2, and the router on the 192.168.10.0/24 subnet.

Successful ping replies between all devices confirm local network connectivity. The accompanying images illustrate the expected configuration; actual success must be verified in Packet Tracer.


Example output:

Step 5: Test Connectivity When the Device Is Offline



4. Observation Table

Parameter	Device Online	Device Offline
Destination IP	192.168.1.5	192.168.1.5
Packets Sent	4	4
Packets Received	4	0
Packet Loss	0%	100%
Status	Successful	Failed

5. What Happens When the Device Is Offline?

When the destination device is offline, it cannot respond to ping requests. The Command Prompt may display **"Request timed out"** or **"Destination host unreachable."** This indicates that the destination is not responding. A firewall or network issue can also cause ping failure.

6. Result

The ping command was used to test connectivity between two devices on the same local network. In the example, the online device responded successfully with 0% packet loss, while the offline device showed 100% packet loss.

7. Conclusion

The ping command is an important network troubleshooting tool used to check connectivity between devices. It helps identify whether a destination device is responding over the network.

8. Screenshots Required

1. Screenshot of the successful ping when the device is online.
2. Screenshot of the failed ping when the device is offline.

Note: The IP address and ping outputs are examples. Replace them with your actual test results before submitting the assignment.